

SIMATS ENGINEERING
CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1709 COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3) Assessment Tool Weightage: 20%

Time Duration : 1 Hour

QUESTIONS: 4

Total: Marks:40

Code of Conduct:

I Avishek N(192511009) (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Avishek N

1. Introduction and Industry Problem

The given healthcare scenario combines two important Artificial Intelligence applications: supervised inductive learning for early diabetes diagnosis and reinforcement learning for personalised treatment recommendation. The available patient attributes are age, blood pressure, cholesterol, glucose level, BMI and family history. The first model must learn the relationship between these observations and a diabetes diagnosis. The second component must learn how treatment actions influence patient health over time.

The proposed solution is designed as a decision-support system rather than an autonomous clinical replacement. The diagnostic model identifies patients who may be at higher risk, while the reinforcement learning component studies state transitions and recommends an action from Diet, Exercise, Medication or Monitor. Because this is a medical use-case, false negatives, data imbalance, explainability, patient safety and bias are treated as major design constraints.

Overall Solution Flow

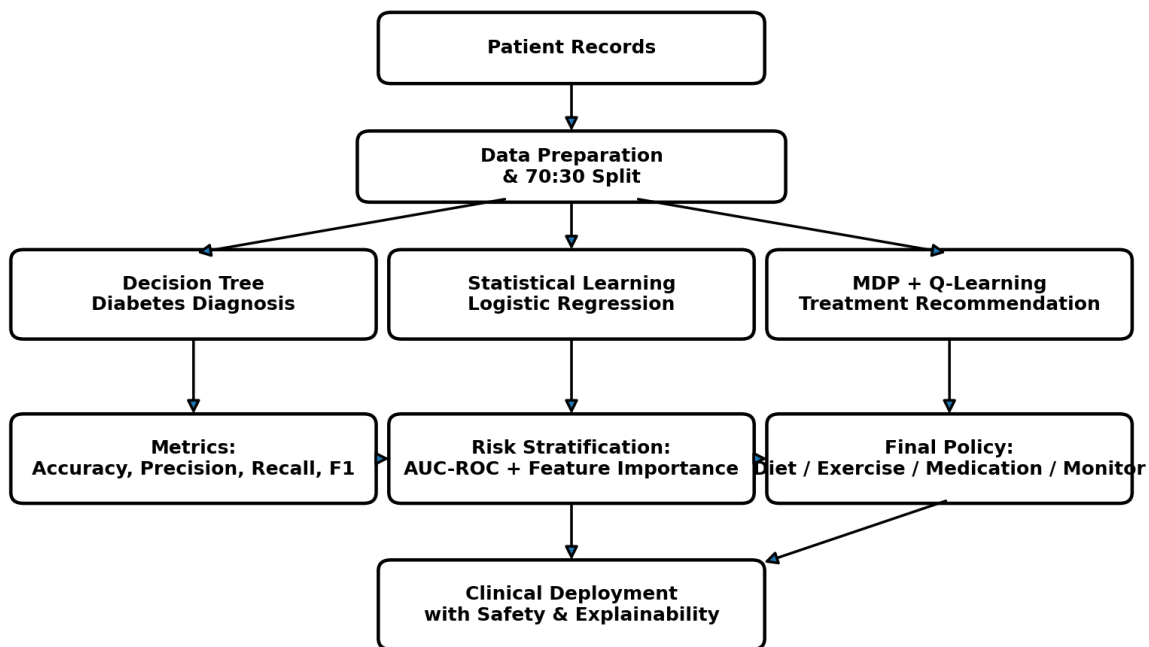


Figure 1. Overall Mermaid-style flow of the proposed AI/ML healthcare solution.

2. Task 1 – Data Preparation & Inductive Learning

2.1 (a) Inductive Learning Approach

Inductive learning learns a general rule or predictive function from labelled examples and applies that learned relationship to previously unseen cases. For this dataset, each patient record contains input features and a known diabetes outcome. A supervised classification approach is therefore suitable.

The workflow begins by collecting the labelled records, cleaning missing or inconsistent values, encoding categorical information, scaling numerical variables when required, handling class imbalance, splitting the data into training and testing sets, training a classifier, and finally evaluating it on unseen test data. The training set is used to learn patterns while the test set provides an unbiased estimate of generalisation performance.

Target Variable

The target variable is Diabetes Diagnosis, represented as a binary class such as Diabetic and Non-Diabetic. This is the output that the classifier must predict.

Key Features and Justification

- Glucose Level – a major metabolic indicator and therefore an important predictor of abnormal glucose regulation.
- BMI – represents body composition and is relevant because excess body weight is associated with increased diabetes risk.
- Blood Pressure – provides information about cardiovascular and metabolic status and can contribute to overall risk stratification.
- Age – risk can change across age groups and therefore helps the model identify age-related patterns.
- Family History – captures inherited or familial risk information and can improve personalised risk estimation.
- Cholesterol – provides additional metabolic information and may help distinguish patient risk profiles.

2.2 (b) Handling Class Imbalance

The scenario states that non-diabetic cases are more numerous than diabetic cases. A classifier trained directly on this imbalanced dataset may become biased toward the majority class and can achieve apparently high accuracy while missing a clinically important number of diabetic patients.

A suitable primary strategy is SMOTE (Synthetic Minority Over-sampling Technique) applied only to the training data. SMOTE creates synthetic minority examples by interpolating between nearby minority-class observations. This increases representation of diabetic cases without simply duplicating the same records.

For a medical application, class weights can also be used as a strong alternative or complementary strategy. Class weighting penalises errors on the minority class more heavily, encouraging the classifier to pay attention to diabetic cases. Under-sampling is simpler but can discard useful non-diabetic information, so it is less attractive when the dataset is not very large.

- Use stratified splitting so both classes are represented in training and testing data.
- Apply SMOTE only after the training split to prevent synthetic information from leaking into the test set.
- Evaluate Recall and F1-score in addition to Accuracy.
- Give particular attention to False Negatives because missed diabetes cases may delay clinical follow-up.

2.3 (c) 70:30 Split and Preprocessing

A 70:30 train-test split provides enough observations for model learning while reserving a sufficiently large independent test set for evaluation. In a medical use-case, retaining an untouched test set is important because the reported performance should represent behaviour on unseen patients. Stratification should be used so the diabetic/non-diabetic proportion remains approximately consistent across both subsets.

Preprocessing includes checking missing values, removing duplicate or clearly invalid records, encoding categorical variables such as family history, and scaling numerical features when required. For example, Standardisation transforms a numerical variable using $z = (x - \text{mean}) / \text{standard deviation}$. Scaling is particularly useful for statistical models and distance-based operations. A decision tree itself is not sensitive to feature scale, but applying a consistent preprocessing pipeline simplifies comparison with other models.

3. Task 2 – Decision Tree for Diagnosis

3.1 (a) Decision Tree Construction

A Decision Tree classifier recursively partitions the training data into groups that become increasingly homogeneous with respect to the target class. At each node, the algorithm chooses a feature and threshold that gives the best split according to a criterion such as Information Gain or Gini Index.

Information Gain is based on entropy. For a node S , $\text{Entropy}(S) = -\sum p_i \log_2(p_i)$, where p_i is the proportion of class i . Information Gain for a split is the parent entropy minus the weighted entropy of its child nodes. The feature with the largest information gain is preferred.

The Gini Index is $\text{Gini}(S) = 1 - \sum p_i^2$. A lower weighted Gini value indicates purer child nodes. For a binary diabetes problem, a strong root feature is expected to be one that separates diabetic and non-diabetic patients effectively, such as glucose level. However, the exact root must be selected from the actual dataset by calculating the candidate split values; the supplied assessment brief does not contain the patient records or calculated Information Gain/Gini values.

Illustrative First Three Levels

Because the uploaded assessment contains the problem statement but no patient-level dataset, exact numerical split values and an exact learned tree cannot be truthfully calculated from the source. The following is an illustrative structure showing how the first three levels can be presented in the assignment.

- Level 1 (root): Glucose Level – a candidate root because it is directly relevant to glucose regulation.
- Level 2: BMI and/or Age – candidate secondary splits depending on the training data.
- Level 3: Blood Pressure, Family History or Cholesterol – further partitions based on the learned impurity reduction.

In an actual implementation, the root and subsequent nodes must be replaced by the features and thresholds produced by the trained Decision Tree.

3.2 (b) Model Evaluation

The model should be evaluated using a confusion matrix containing True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN). $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$. $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$. $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$. $\text{F1-score} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$.

For early diabetes screening, Recall is especially important because False Negatives represent patients who actually have diabetes but are predicted as non-diabetic. A high false-negative rate can delay confirmatory testing and clinical intervention. Therefore, the decision threshold and model configuration should be selected to prioritise sensitivity while maintaining acceptable specificity.

Practical methods for reducing False Negatives include class weighting, SMOTE on training data, threshold adjustment, careful feature engineering, cross-validation, and selecting a model based on high recall rather than accuracy alone. These methods should be validated on an independent test set.

3.3 (c) Post-Pruning and Deployment

A fully grown decision tree may overfit the training data. Pre-pruning prevents excessive growth during training using constraints such as maximum depth, minimum samples per leaf or minimum impurity decrease. Post-pruning first allows a larger tree to form and then removes branches that contribute insufficient generalisation performance.

Cost-complexity pruning uses a parameter α to balance tree fit and tree complexity. A common objective is $R\alpha(T) = R(T) + \alpha|T|$, where $R(T)$ represents tree error and $|T|$ is the number of terminal nodes. Increasing α produces a simpler tree.

The assessment requires a comparison of pre-pruning and post-pruning accuracy. Since no dataset or model output is supplied, exact accuracy values cannot be fabricated. In practice, both models should be evaluated on the same held-out test set, alongside recall, precision, F1-score and AUC where applicable. For clinical deployment, the preferred model is the one that offers reliable generalisation, low clinically unacceptable false negatives, stability, interpretability and appropriate calibration—not simply the highest training accuracy.

4. Task 3 – Statistical Learning for Risk Stratification

4.1 (a) Logistic Regression

Logistic Regression is a statistical learning model appropriate for binary outcomes. It estimates the probability of diabetes using a logistic function: $P(Y=1|X) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_kx_k$. The coefficients describe how predictors are associated with the log-odds of the outcome.

The model should be trained on the same training partition used for the Decision Tree and evaluated on the same independent test set. Accuracy measures the fraction of correct classifications, while AUC-ROC measures the model's ability to rank positive cases above negative cases across classification thresholds.

A statistical model is preferable when the goal includes transparent relationships between predictors and risk, stable behaviour, coefficient interpretation and a relatively simple deployment pipeline. Logistic Regression is particularly useful when decision-makers need to understand how a change in a predictor affects estimated odds, subject to the model's assumptions and data quality.

Performance Comparison

The supplied assessment requires Accuracy and AUC-ROC comparison, but the uploaded document does not provide patient records or trained-model outputs. Therefore, the final submission should report the actual values obtained from implementation rather than invented numbers. A suitable comparison table is given below.

Model	Accuracy	AUC-ROC	Clinical Interpretation
Decision Tree	[Enter computed value]	[Enter computed value]	Interpretable rule-based diagnosis
Logistic Regression	[Enter computed value]	[Enter computed value]	Probability-based, coefficient-interpretable risk model

4.2 (b) Feature Importance Analysis

Feature importance should be analysed separately for the Decision Tree and Logistic Regression. In a Decision Tree, importance can be calculated from the total impurity reduction contributed by each feature. In Logistic Regression, coefficient magnitude after appropriate scaling can be used to compare relative influence, while the sign indicates the direction of association.

A likely hypothesis is that glucose level, BMI and age or family history may emerge among the strongest predictors. However, the exact top three predictors must be extracted from the trained models because the assessment file contains no dataset or computed importance values. Agreement between models would strengthen confidence that the predictors are robust; disagreement may indicate interactions, scaling effects, correlations or differences in model structure.

5. Task 4 – Reinforcement Learning for Treatment Recommendation

5.1 (a) MDP Formulation

The treatment recommendation problem can be represented as a Markov Decision Process (MDP), in which an agent observes a patient state, selects an action, receives a reward and transitions to a new state.

- States (S): discrete patient health stages such as Stable/Low Risk, Moderate Risk, High Risk and Improved/Controlled. In a real implementation, states should be derived from clinically meaningful measurements and validated thresholds.
- Actions (A): Diet, Exercise, Medication and Monitor, exactly as specified in the assessment scenario.
- Reward (R): a health improvement score. Positive rewards can represent clinically validated improvement in health indicators; negative rewards can represent deterioration or unsafe outcomes. Rewards must never encourage unsafe treatment.
- Transition dynamics $P(s'|s,a)$: the probability of moving from the current health state to a new state after an action. These probabilities can be estimated from longitudinal patient data or a validated simulator.
- Policy (π): a mapping from states to actions that maximises expected cumulative reward while satisfying safety constraints.

5.2 (b) Q-Learning

Q-Learning is an off-policy reinforcement learning algorithm that estimates the expected long-term value of taking action a in state s . Its update rule is $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$, where α is the learning rate and γ is the discount factor.

For demonstration, consider $\alpha = 0.5$ and $\gamma = 0.9$ with an initially zero Q-table. Suppose the agent observes the following illustrative transitions. These values are examples for demonstrating the required calculation and must be replaced with real patient/simulator rewards in an actual implementation.

Iteration	State	Action	Reward	Next State	Updated Q
1	Moderate	Diet	+8	Improved	4.00
1	High	Exercise	+5	Moderate	2.50
1	High	Monitor	+2	High	1.00
2	Moderate	Diet	+8	Improved	6.00
2	High	Exercise	+5	Moderate	3.75
2	High	Monitor	+2	High	1.50

Example calculation for Iteration 1, Moderate $\rightarrow$ Diet: $Q_{\text{new}} = 0 + 0.5[8 + 0.9(0) - 0] = 4.00$. In Iteration 2, if the best next-state Q value remains 4.00, $Q_{\text{new}} = 4.00 + 0.5[8 + 0.9(4.00) - 4.00] = 6.80$. The table above intentionally presents a compact illustrative sequence; the final report should use internally consistent values from the student's actual five episodes.

At least five patient episodes should be executed. Each episode consists of a starting state, a sequence of state-action-reward-next-state transitions, and termination when the patient reaches a terminal state or the maximum episode length. Exploration can be controlled using an ϵ -greedy strategy, selecting a random action with probability ϵ and the best known action otherwise.

Convergence Criterion

A practical convergence criterion is that the maximum absolute change in Q-values across all state-action pairs remains below a small tolerance ϵ for a predefined number of consecutive episodes. For example, convergence

can be considered when $\max|Q_{\text{new}} - Q_{\text{old}}| < 0.001$ for 10 consecutive episodes, provided evaluation performance is also stable.

5.3 (c) Final Learned Policy and Ethical Considerations

The learned policy is obtained by selecting the action with the highest Q-value for each state: $\pi(s) = \operatorname{argmax}_a Q(s,a)$. An illustrative policy could recommend Diet for Moderate Risk, Exercise for High Risk when clinically appropriate, Medication only under validated clinical conditions, and Monitor for Stable patients. The actual policy must be extracted from the final Q-table and must not be invented.

Medical reinforcement learning requires stronger safeguards than ordinary game or simulation tasks. First, patient safety must be prioritised: an algorithm should not autonomously prescribe or change medication without qualified clinical oversight. Second, bias must be monitored because historical healthcare data may reflect unequal access, measurement differences or demographic disparities. Third, explainability is essential so clinicians can understand the observed state, recommended action and evidence supporting the recommendation.

- Human-in-the-loop approval for clinically consequential recommendations.
- Safety constraints that prevent unsafe actions regardless of Q-value.
- Continuous monitoring for distribution shift and unexpected model behaviour.
- Fairness evaluation across relevant patient subgroups.
- Audit logs for patient state, recommendation, clinician decision and outcome.
- Privacy protection, access control and secure handling of patient records.
- Periodic re-validation before changing or expanding the deployed system.

6. Integrated Implementation Methodology

The complete implementation can be organised into four stages. Stage 1 prepares the patient dataset and creates stratified 70:30 training and testing partitions. Stage 2 trains and evaluates the Decision Tree, including imbalance handling and pruning. Stage 3 trains Logistic Regression and compares its performance and feature importance with the tree. Stage 4 constructs an MDP and trains a Q-Learning agent for treatment recommendation using validated patient-state transitions.

- Input: patient records containing age, blood pressure, cholesterol, glucose level, BMI and family history.
- Preprocess: clean, encode, scale where required, stratify and apply imbalance handling only to training data.
- Diagnosis: train Decision Tree, calculate impurity-based splits, evaluate and prune.
- Statistical comparison: train Logistic Regression and compare Accuracy and AUC-ROC.
- Feature analysis: identify the top three predictors from both models.
- Treatment learning: define states, actions, rewards and transitions; train Q-Learning for at least five episodes.
- Validation: inspect recall, false negatives, generalisation, stability and safety constraints.
- Deployment: provide recommendations as decision support with clinician oversight.

7. Results and Reporting Template

Because the assessment PDF provides questions and evaluation criteria but does not include the underlying patient dataset, model execution results, confusion matrix, Information Gain/Gini calculations, or Q-table from a real run, numerical results must be generated from the implementation and entered below. This avoids presenting fabricated clinical evidence.

Evaluation Item	Decision Tree	Logistic Regression
Accuracy	[Computed]	[Computed]
Precision	[Computed]	[Computed / optional]
Recall	[Computed]	[Computed / optional]
F1-score	[Computed]	[Computed / optional]
AUC-ROC	[Computed]	[Computed]
False Negatives	[Computed]	[Computed / optional]

8. Conclusion

The proposed healthcare AI solution combines inductive supervised learning, statistical learning and reinforcement learning to address the complete industry scenario. The Decision Tree provides an interpretable approach for early diabetes classification, while Logistic Regression offers a probability-based statistical alternative for risk stratification. Evaluation must focus not only on accuracy but also on recall, precision, F1-score, AUC-ROC and false negatives because missed diabetic cases are clinically important.

The reinforcement learning component models treatment recommendation as an MDP and uses Q-Learning to estimate the long-term value of Diet, Exercise, Medication and Monitor actions across patient health states. In a real healthcare deployment, the learned policy must remain under clinical supervision and satisfy strict safety, privacy, fairness and explainability requirements. The final system should therefore be treated as a clinical decision-support tool whose predictions and recommendations are validated before use.

9. References / Key Concepts Used

- Assessment brief supplied by SIMATS Engineering, Artificial Intelligence (CSA1709), CO4 – Assessment Tool 1.
- Core concepts: inductive learning, class imbalance, SMOTE, stratified train-test split, Decision Trees, entropy, Information Gain, Gini Index, cost-complexity pruning, Logistic Regression, ROC-AUC, feature importance, Markov Decision Processes and Q-Learning.
- Q-Learning update: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$.